

Recall from last class:

- If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v , called an **eigenvector** corresponding to λ , such that

$$Av = \lambda v.$$

- If λ is an eigenvalue of an $n \times n$ matrix A , then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ .
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \dots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_p$ of an $n \times n$ matrix A , then $v_1, \dots, v_p$ are linearly independent.

- If A is an $n \times n$ matrix, then the characteristic polynomial of A , denoted by $p_A(\lambda)$, is given by

$$p_A(\lambda) = \det(A - \lambda I_n).$$

- λ is an eigenvalue of an $n \times n$ matrix A if and only if λ is a zero of the characteristic polynomial.
- If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).
- An $n \times n$ matrix A is **diagonalizable** if A is similar to a diagonal matrix.
- **Diagonalization Theorem:** An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A . In this case the diagonal entries of D are the corresponding eigenvalues of A .
- **Theorem 6:** An $n \times n$ matrix A with n distinct eigenvalues is diagonalizable.
- **Theorem 7:** Let A be an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \dots, \lambda_p$.

1. For $1 \leq k \leq p$, the dimension of the eigenspace of λ_k is at most the algebraic multiplicity of the eigenvalue λ_k .
2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces is n if and only if the characteristic polynomial factors into linear factors and the dimension for each eigenspace is exactly the algebraic multiplicity.
3. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k , then $\mathcal{B}_1, \dots, \mathcal{B}_p$ is an eigenbasis for $\mathbb{R}^n$.

1. Let $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Find eigenvalues and eigenspaces.

2. Let A be an $n \times n$ matrix with real entries. Suppose λ is an eigenvalue of A , then is $\bar{\lambda}$ an eigenvalue of A .

3. Let $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ where a, b are real and not both zero, then find the eigenvalues of C .

Theorem: Let A be a real 2 by 2 matrix with a complex eigenvalue $\lambda = a - bi$ ($b \neq 0$) and an associated eigenvector v in $\mathbb{C}^2$. Then $A = PCP^{-1}$ where $P = [\text{Re}(v) \text{ Im}(v)]$ and

$$C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}.$$

4. Prove the theorem.

5. Let $A = \begin{bmatrix} 3 & -2 \\ 4 & -1 \end{bmatrix}$, then find P and C such that $A = PCP^{-1}$ as in the previous theorem.